

E-Commerce Customer Behavior Analysis

End-to-End Data Analytics Project Report

Tools: Python (Pandas) • PostgreSQL • Power BI

Dataset: 5,000 customer records • 30 original attributes

This report documents the analytical workflow used to understand customer profile, spending, purchase behavior, conversion activity, acquisition channels, returns, loyalty and churn. The analysis combines Python-based exploratory analysis, PostgreSQL querying and an interactive Power BI dashboard.

Project Type: Portfolio / Business Analytics Case Study

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1. Executive Summary

The project analyzes a synthetic Indian e-commerce customer dataset containing **5,000 customers**. Across the dataset, customers generated approximately **₹4.05 crore** in recorded spending across **26,020 orders**. The analysis was designed to move beyond descriptive charts and identify customer segments, behavioral patterns, conversion opportunities, acquisition performance, return behavior and churn risk.

The strongest portfolio-level findings are the concentration of customers and spending in the 26–35 age group, the large gap between product views and completed orders, the importance of Organic Search for acquisition revenue, the high overall churn level of **38.5%**, and the higher return intensity observed in selected categories.

The project also identifies data-quality considerations, particularly the large number of records with missing membership tier information. These issues are explicitly separated from genuine behavioral findings so that the business conclusions are not overstated.

Overall project flow

Raw Dataset → Data Cleaning → Python EDA → PostgreSQL Analysis → Power BI Dashboard → Insights → Recommendations

2. Problem Statement

An e-commerce business generates customer-level information across demographics, transactions, digital engagement, marketing acquisition, payments, membership, returns and churn. However, having these fields does not automatically provide management with an understanding of which customer groups generate value, where the purchase funnel loses customers, which channels perform effectively, or which segments require retention attention.

The central problem addressed by this project is therefore:

How can customer-level e-commerce data be analyzed to identify the customer segments, purchasing behaviors, conversion patterns, acquisition channels, return patterns and churn characteristics that can support better commercial and customer-retention decisions?

Sub-problems addressed

- Identify the main customer segments by age, gender, geography, membership and loyalty.
- Understand which segments and product categories contribute the most spending and orders.
- Examine whether purchase frequency, engagement and discount usage are associated with customer spending or ordering behavior.
- Measure the movement from product views to cart additions, checkout starts and completed orders.
- Compare acquisition channels by customer volume, revenue and conversion-related measures.
- Assess payment and delivery preferences to understand customer usage patterns.
- Measure return behavior and identify categories with relatively high return intensity.
- Measure overall churn and compare churn across loyalty and membership groups.
- Use SQL to perform grouped, conditional, subquery and window-function analysis on customer records.
- Present the results in an interactive Power BI dashboard for executive-level exploration.

3. Business Objectives & Analytical Questions

The project objectives were designed to create a continuous analytical story rather than a collection of unrelated charts.

Objective	Key analytical questions
Customer profile	Who are the customers by age, gender, state, membership and loyalty?
Customer value	Which groups and categories generate the most spending? Does order frequency correspond with higher spending?
Product performance	Which categories are most represented, generate the most revenue, and show higher return intensity?
Customer engagement	Do higher product views lead to more cart additions and do more cart additions lead to more checkout starts?
Conversion	Where is the customer funnel concentrated or lost, and how does conversion vary by device and acquisition channel?
Acquisition	Which channels bring the most customers and revenue, and which channels bring higher-spending customers?
Payments & delivery	Which payment methods and delivery options are most commonly used?
Retention	How large is churn, and which loyalty/membership groups show greater churn exposure?
Data quality	Which fields contain missing information and how should missing values be handled without inventing unsupported b

4. Dataset Overview

The project uses a synthetic Indian e-commerce customer dataset. Each row represents one customer record and contains demographic, transactional, engagement, marketing and retention attributes.

Dataset characteristic	Value
Records	5,000 customer records
Original columns	30
Age range	18–58 years
Total recorded spending	■40,537,703.34
Total recorded orders	26,020
Total product views	418,750
Total cart adds	60,163
Total checkout starts	30,730
Average customer-level AOV field	■1,553.85
Total spend / total orders	■1,557.94

Main variable groups

- **Customer profile:** Customer ID, name, age, gender, state, city.
- **Transaction behavior:** orders count, total spend, average order value, discount amount, returns.
- **Digital engagement:** sessions, product views, cart adds, checkout starts, conversion rate.
- **Commercial context:** product category, payment method, device type, acquisition channel, membership tier, delivery type.
- **Customer experience and retention:** support contacts, average rating, loyalty segment and churn status.

5. Data Preparation & Quality Handling

Data preparation was performed in Pandas before the dataset was exported for SQL analysis and used as the analytical source for the Power BI dashboard. The cleaning steps followed the logic documented in the submitted notebook.

5.1 Initial quality assessment

Field	Missing records	Handling
Discount Amount	50	Set to 0 when Discount Used = No; remaining missing values for discount users were filled with 0.
Membership Tier	2,205	Replaced missing values with the explicit category 'Unknown' rather than assuming a membership tier.
Support Contacts	75	Filled with the median value of 1 and converted the field to integer.
Average Rating	175	Missing values remained missing; no unsupported imputation was applied in the notebook.

5.2 Transformations

- Column names were standardized to lowercase, underscore-separated names.
- Signup Date and Last Order Date were converted to datetime format using day-first parsing.
- Age groups were created as 18–25, 26–35, 36–45, 46–55 and 56+.
- Product-view groups were created as Low, Medium and High using the notebook's specified cut points.
- The cleaned dataset was exported for PostgreSQL analysis.

5.3 Data-quality interpretation

The 2,205 Unknown membership records are analytically important because membership comparisons can be distorted if the missing category is treated as a genuine membership tier. The report therefore treats Unknown as a data-quality category rather than interpreting it as a customer segment with a business meaning.

6. Exploratory Data Analysis

Python/Pandas was used to establish the customer and behavioral patterns before the dashboard was built.

6.1 Customer profile

Measure	Result
Largest age group	26–35: 2,365 customers
Second-largest age group	36–45: 1,255 customers
Largest customer state	Uttar Pradesh: 459 customers
Gender distribution	Male 2,480; Female 2,400; Other 120
Known membership tiers	Silver 1,388; Gold 1,044; Platinum 363
Largest loyalty segment	Medium: 3,847 customers

6.2 Spending and product performance

Analysis	Result
Largest spending age group	26–35: ■1.94 crore (47.8% of total spending)
Most represented category	Fashion: 1,009 customer records
Highest total category revenue	Fashion: ■8,355,819
Highest average customer spending by category	Books: ■8,380
Highest observed AOV customer	■8,646.44
Discount users	3,131 customers; non-users 1,869

6.3 Engagement and conversion

Analysis	Result
Product views	418,750 total
Cart adds	60,163 total
Checkout starts	30,730 total
Orders	26,020 total
Views-to-orders funnel conversion	6.2%
Low/Medium/High view groups	1,263 / 2,555 / 1,182 customers
Average cart adds	5.92 for Low-view, 11.71 for Medium-view, 19.26 for High-view customers

6.4 Acquisition, payments and delivery

Analysis	Result
Largest acquisition channel by customer count	Organic Search: 1,236
Largest acquisition channel by revenue	Organic Search: ₹0.99 crore
Highest average customer spend by acquisition channel	Referral: ₹8,518
Highest mean conversion rate by acquisition channel	YouTube: 17.76%
Most-used payment method	UPI: 2,025 customers
Most-used delivery type	Standard: 3,393 customers

6.5 Returns and retention

Analysis	Result
Customers with zero returns	3,487
Total recorded returns	1,780
Average customer return rate field	8.26%
Highest return intensity by category	Mobile Accessories: 7.58% of orders
Overall churned customers	1,925
Overall churn rate	38.5%
Highest loyalty-segment churn rate	Low loyalty: 45.0%
Highest average spending loyalty segment	High loyalty: ₹23,003

Important analytical distinction

The dataset contains a customer-level *Conversion Rate* field whose average is around 16–17% across devices. Separately, the dashboard funnel compares total product views with total orders, producing a views-to-orders ratio of 6.2%. These are different measures and should not be presented as the same KPI.

7. SQL Analysis

The PostgreSQL script converts the cleaned customer dataset into a relational table and then performs 24 analytical queries. The SQL work complements the Pandas exploration by answering business questions using aggregation, conditional logic, subqueries and window functions.

7.1 SQL analysis areas

Area	Examples from the submitted SQL script
Category performance	Revenue and order totals by product category.
Loyalty analysis	Customer count, total spending and average customer spending by loyalty segment.
Payments & geography	Payment-method usage and state-level customer, order and revenue analysis.
Customer targeting	High-value customers, high-frequency/low-spending customers, no-return customers and highly engaged low-co
Conditional aggregation	Discount-user counts and discount usage percentages by category.
Customer value segmentation	CASE-based High / Medium / Low value classification.
Subquery analysis	Customers spending above the overall average.
Window functions	ROW_NUMBER, RANK and LAG for state-level ranking and customer-to-customer comparisons.
Retention	Churn rate by membership tier and identification of churned customers with high product engagement.

7.2 Examples of SQL techniques demonstrated

- **GROUP BY + aggregate functions:** SUM, COUNT and AVG for commercial summaries.
- **HAVING:** filtering grouped categories using revenue/order conditions.
- **CASE:** creating customer-value classifications and conditional counts.
- **Subqueries:** comparing individual customer spending with the overall average.
- **Window functions:** ROW_NUMBER and RANK for within-state customer ranking; LAG for previous-customer comparison ordered by signup date.
- **PostgreSQL casting:** numeric division using `::numeric` where required.

7.3 Representative SQL-derived findings

- Fashion leads category revenue at approximately ₹0.84 crore.
- High-loyalty customers have average spending of approximately ₹23,003, versus ₹8,401 for Medium and ₹3,258 for Low loyalty.
- UPI is the most-used payment method in the customer dataset.
- Uttar Pradesh is the highest-revenue state in the state-level analysis.
- The SQL script also identifies targeted customer groups such as high-value customers and highly engaged customers with low conversion.

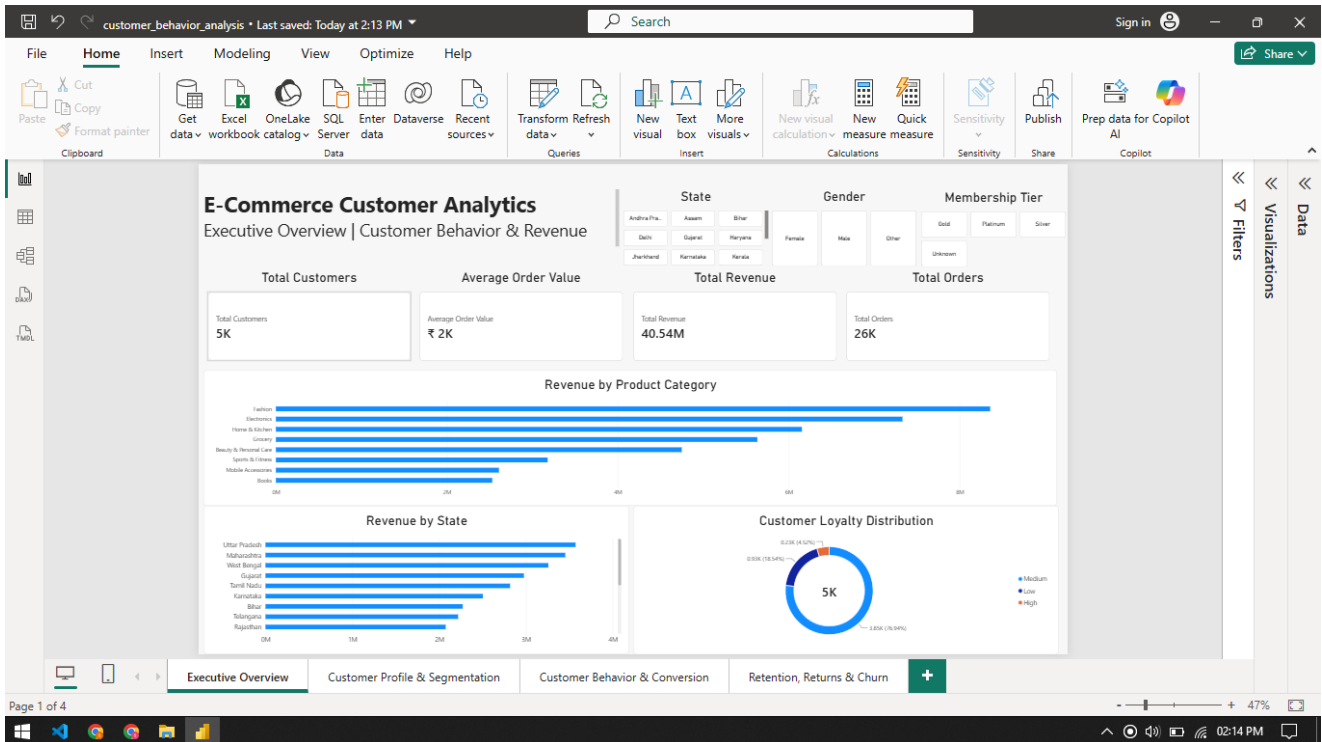
8. Power BI Dashboard

The Power BI report turns the analysis into a four-page interactive dashboard. The pages are arranged from executive summary to customer segmentation, behavior/conversion and finally retention/returns/churn.

8.1 Dashboard architecture

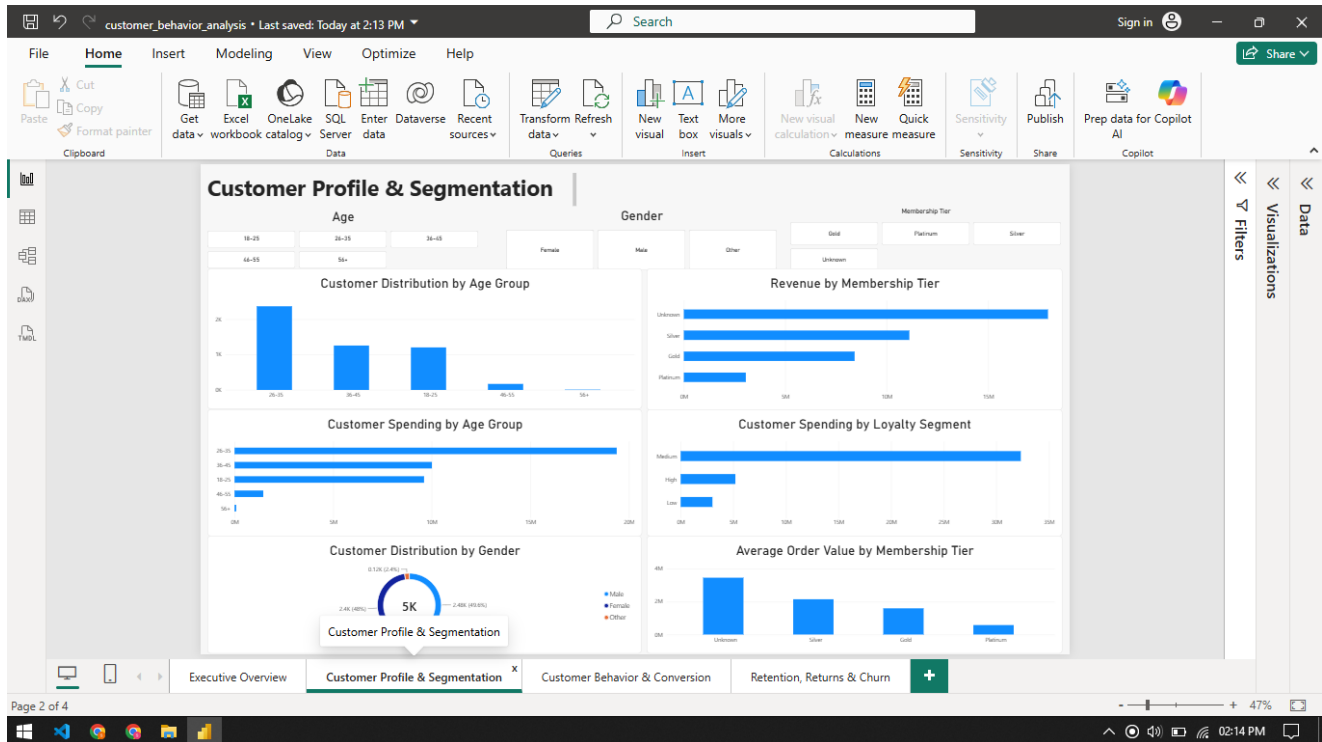
Page	Purpose	Main visuals
Executive Overview	High-level commercial and customer snapshot.	KPI cards, revenue by product category, revenue by state, loyalty distribution
Customer Profile & Segmentation	Understand who the customers are and how value they bring.	Age distribution, membership revenue, age-group spending, loyalty spend
Customer Behavior & Conversion	Understand digital engagement and the conversion process.	Product view/cart/checkout KPIs, funnel, acquisition revenue, device comparison
Retention, Returns & Churn	Identify retention risk and return patterns.	Churn KPIs, churn by loyalty, return rate by category, churn by membership

8.2. Executive Overview



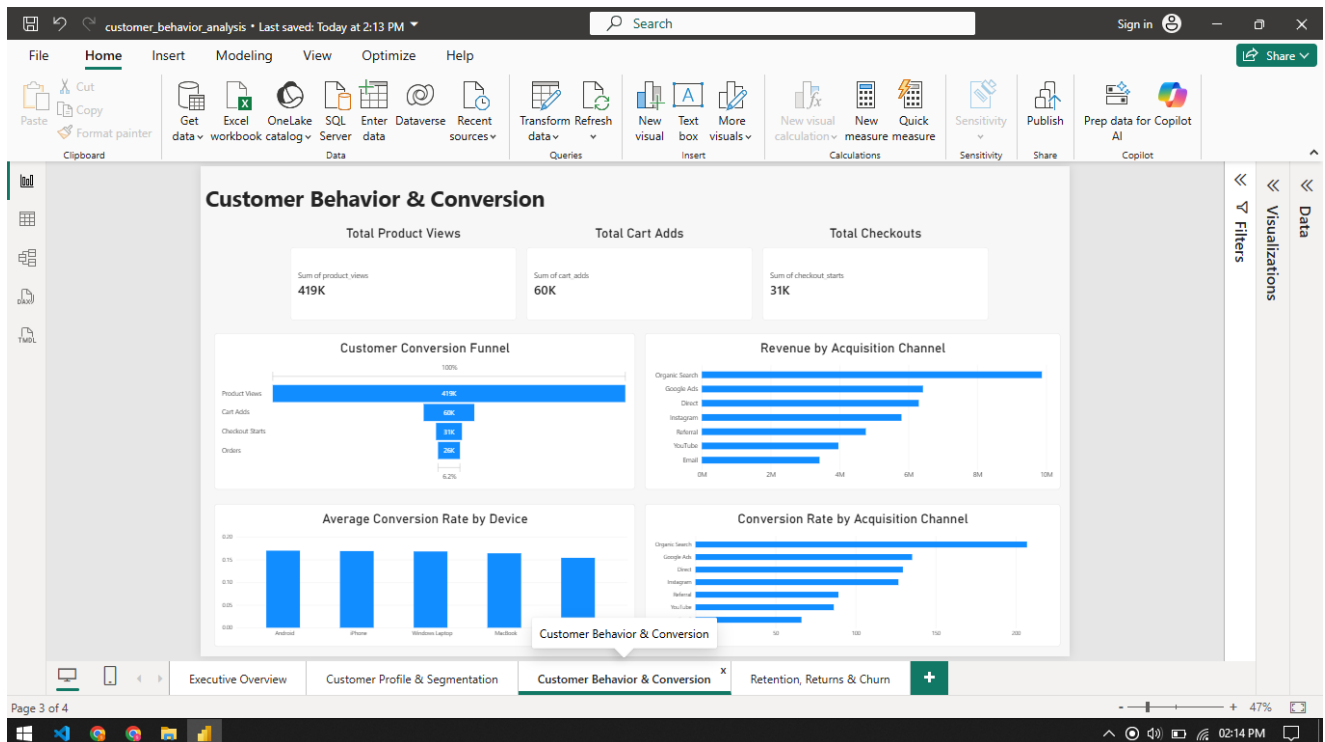
Dashboard page shown above: Executive Overview.

8.3. Customer Profile & Segmentation



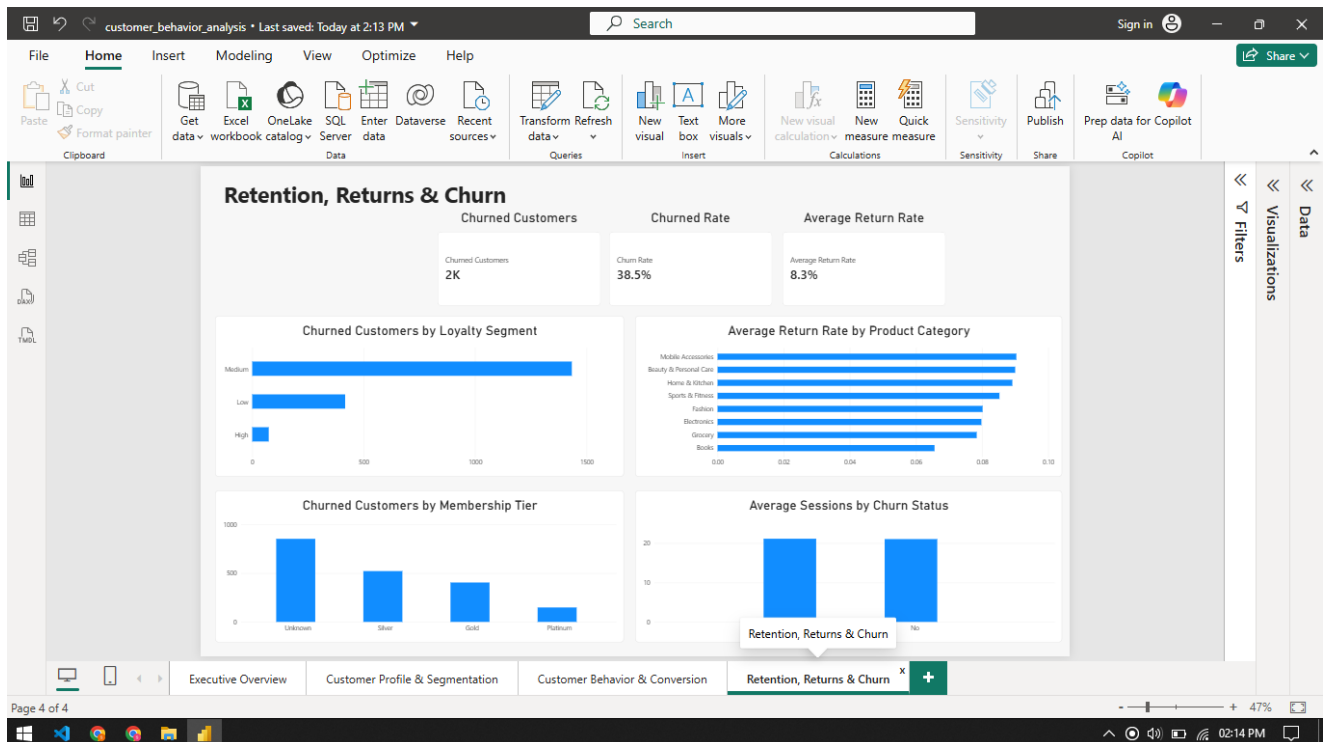
Dashboard page shown above: Customer Profile & Segmentation.

8.4. Customer Behavior & Conversion



Dashboard page shown above: Customer Behavior & Conversion.

8.5. Retention, Returns & Churn



Dashboard page shown above: Retention, Returns & Churn.

9. Key Business Insights

The following conclusions are selected from the combined Python, SQL and Power BI analysis. Each insight is stated with the evidence that supports it and avoids treating descriptive counts as causal relationships.

9.1 The 26–35 segment is the central customer group

The 26–35 age group contains **2,365 customers** and accounts for approximately **47.8%** of total customer spending. Its average customer spending is about **₹8,195**.

9.2 The purchase funnel has a substantial views-to-orders gap

The dataset records 418,750 product views but 26,020 orders, giving a total views-to-orders ratio of approximately **6.2%**. Engagement increases meaningfully across view groups: high-view customers average 19.26 cart adds compared with 5.92 among low-view customers.

9.3 Organic Search is the largest acquisition contributor by volume and revenue

Organic Search brings 1,236 customers and generates approximately **₹0.99 crore**, the highest revenue among the listed acquisition channels. This makes it an important channel for both customer acquisition scale and recorded revenue contribution.

9.4 Customer retention is a major area of concern

There are **1,925 churned customers**, corresponding to a churn rate of **38.5%**. Low-loyalty customers show the highest churn rate at approximately 45.0%, compared with 37.3% for Medium and 32.7% for High loyalty.

9.5 Return intensity is higher in selected categories

Mobile Accessories has the highest return rate calculated from total returns divided by total orders (7.58%), followed by Beauty & Personal Care (7.42%). Books has the lowest category return rate at 5.35%.

9.6 Loyalty level is strongly associated with customer spending

High-loyalty customers average approximately **₹23,003** in spending, versus **₹8,401** for Medium and **₹3,258** for Low loyalty. This is a strong segmentation signal in the dataset, although the analysis does not establish causality.

9.7 Membership analysis is constrained by missing classification

2,205 records, or approximately 44.1% of customers, have Unknown membership tier after cleaning. Because this is a large share of the dataset, membership-level comparisons should be interpreted cautiously.

9.8 Discount usage does not show higher order frequency

Customers who used discounts placed an average of about 5.19 orders, while non-users averaged about 5.22 orders. Within this dataset, discount usage therefore does not appear to be associated with a higher order count.

10. Business Recommendations

Recommendations below translate the observed patterns into possible business actions. They are framed as actions to investigate or test rather than claims that the dataset has proven a causal effect.

Observed evidence	Recommended action
26–35 accounts for 47.8% of spending.	Prioritize this segment for personalized campaigns, loyalty offers and retention experiments where appropriate.
Views-to-orders ratio is 6.2%.	Investigate product-page quality, pricing, product relevance, cart abandonment and checkout friction.
Organic Search leads customer volume and revenue.	Protect and expand high-performing SEO/content activity and evaluate whether additional organic search is warranted.
Overall churn is 38.5%; Low loyalty has the highest churn rate.	Develop targeted re-engagement journeys for low-loyalty customers and test whether personalized offers are effective.
Mobile Accessories and Beauty & Personal Care have highest return rates.	Review product descriptions, quality issues, compatibility/sizing information and delivery-related issues.
2,205 customers have Unknown membership.	Improve membership data capture and classification so tier-level customer-value and retention strategies can be targeted.
High-loyalty customers have much higher average spending.	Protect the high-loyalty segment with retention benefits while designing progression paths for Mids and Low loyalty.
Discount users do not place more orders on average.	Avoid assuming that discounting automatically increases purchase frequency; evaluate discount effectiveness.

11. Limitations & Interpretation Notes

- The dataset is synthetic and should be treated as a portfolio case study rather than evidence about a real e-commerce company.
- The analysis is customer-level and largely cross-sectional; it identifies associations and patterns but does not prove causality.
- Membership tier is missing for 2,205 customers, so membership comparisons have a substantial Unknown category.
- Average Rating contains 175 missing values that were not imputed in the submitted notebook.
- The dashboard's views-to-orders funnel ratio and the row-level Conversion Rate field are different metrics and should not be conflated.
- The notebook's session-count analysis uses exact session counts, which can create noisy averages for rare session values; it is more appropriate for exploration than strong business inference.
- Return-rate interpretations depend on the denominator. The report distinguishes the customer-level Return Rate field from the category-level calculation of total returns divided by total orders.
- The project does not include customer-level order timestamps or individual order-line data, so cohort, monthly retention and detailed time-series analyses were outside the submitted dataset structure.

Metric definitions used in this report

Metric	Definition / interpretation
Total Revenue	Sum of Total Spend (INR) across customer records.
Total Orders	Sum of Orders Count across customer records.
Views-to-orders conversion	Total Orders ÷ Total Product Views.
Average customer spending	Mean of Total Spend (INR) across customers in a group.
Category return rate	Total Return Count ÷ Total Orders within the category.
Churn rate	Churned customers ÷ total customers.

12. Conclusion

This project demonstrates an end-to-end customer analytics workflow using a common business dataset and three analytical tools. Pandas was used to inspect, clean and explore the data; PostgreSQL was used to structure and query the dataset through progressively more advanced SQL techniques; and Power BI was used to convert the analytical results into an interactive dashboard.

The analysis indicates that the 26–35 segment is central to current customer value, the digital funnel contains a large gap between browsing and completed orders, Organic Search is the strongest acquisition contributor by recorded revenue, loyalty level is strongly differentiated by spending, and churn represents a substantial retention challenge. Return behavior also varies by product category, while missing membership classification limits some segmentation conclusions.

The project therefore moves from descriptive reporting toward decision support: identifying where customer value is concentrated, where the purchase journey may be leaking, which customer groups need attention, and which operational areas warrant further investigation. The recommendations provide a practical starting point for targeted experiments rather than treating the observed correlations as causal proof.

End of Report